UNIVERSITY OF WOLLONGONG IN DUBAI
Master's in Applied Artificial Intelligence
MAAI901 – Deep Learning for Visual Data

Data-Efficient Vision Transformers for Skin Lesion Classification under Limited and Imbalanced Training Data

Project Report

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Abstract

Deep learning models for skin lesion classification are usually trained on datasets that are both small and strongly imbalanced, so a high overall accuracy can hide poor performance on rare diagnoses. This project investigates whether a Vision Transformer (ViT) that is first pre-trained without labels by masked-patch reconstruction keeps its macro-F1 better than a conventionally trained ViT and a convolutional baseline when only a fraction of the labels is available. A complete, reproducible pipeline was implemented in PyTorch: lesion-grouped data splits, label-budget experiments, self-supervised pre-training, fine-tuning, class-wise evaluation, and automatic report generation.

Experimental results are not included yet. Run python run_all.py to execute the experiments and regenerate this report with real measured numbers.

1. Introduction

Skin lesion classification from dermoscopic images can support clinical decision-making, but labelled medical images are costly to obtain and datasets contain many examples of common diagnoses and few of rare ones. A model can therefore reach a high overall accuracy while missing minority classes, which is the situation where errors matter most. Vision Transformers (Dosovitskiy et al., 2021) model relations between image patches with self-attention, but they have weaker built-in image priors than convolutional networks and are known to need considerable data (Liu et al., 2021).

This project studies the combined effect of limited labels and class imbalance. Rather than claiming that one architecture is universally better, it asks a narrow, testable question and evaluates it with class-sensitive metrics. The system is a research prototype and not a diagnostic tool.

2. Related Work

Liu et al. (2021) showed that ViTs behave differently from CNNs on small datasets and proposed Dense Relative Localization (DRLoc), an auxiliary self-supervised task that encourages spatial reasoning. The study uses general computer-vision data, so its benefit for dermoscopy and imbalanced classes is not established.

Xiao et al. (2023) pre-trained ViTs with masked autoencoders on 266,340 unlabelled chest X-rays and obtained results competitive with DenseNet-121 on thoracic disease classification. The finding supports medical-image self-supervised pre-training but concerns another modality and relies on a large unlabelled collection.

Ansari et al. (2024) addressed class imbalance and skin-tone bias in lesion classification with MixUp-based augmentation for minority classes and an adaptive sampler, evaluated on ISIC

2018 and ASAN. Imbalance and skin-tone bias are related but distinct problems, and the evaluation covered a limited set of model configurations.

Rahmat et al. (2024) combined scaled dot-product, multiplicative and additive attention in a ViT for skin lesion classification on ISIC 2017 and report an F1 improvement of more than 12% over their baseline. Robustness when training data are strongly reduced was not examined.

Shajimon et al. (2023) modified a ViT-B16 with extra layers and regularisation, used augmentation, focal loss and five-fold cross-validation on ISIC 2017, and obtained competitive results against CNN-based methods for a binary task. Broader multi-class settings and darker skin tones remain open.

Synthesis. The five studies address data efficiency, medical pre-training, imbalance and attention design separately. None of them evaluates, for multi-class dermoscopy, how a self-supervised ViT behaves as the number of labels shrinks while minority-class performance is tracked. That intersection is the gap addressed here; the claim is deliberately limited, as other studies may touch parts of it.

3. Problem Statement and Hypothesis

Problem statement: It is unclear how reliably Vision Transformer classifiers recognise minority diagnostic classes of skin lesions when both the labelled training data and the minority-class examples are limited. This project evaluates whether masked-patch self-supervised pre-training improves class-wise reliability under such conditions.

H1: With a limited labelled-data budget, a ViT initialised by masked-patch pre-training achieves a higher test macro-F1 than the same ViT trained from scratch on a multi-class, imbalanced skin lesion dataset.

H0: Under the same budget, the pre-trained ViT does not achieve a higher macro-F1 than the ViT trained from scratch.

4. Dataset

The target dataset is HAM10000 (Tschandl et al., 2018): 10,015 dermatoscopic images in seven imbalanced diagnostic categories. A synthetic proxy can be used for a quick pipeline check.

5. Methodology

5.1 Pre-processing and splits

Images are resized to a square resolution and scaled to [-1, 1]. Data are divided into train, validation and test sets (about 71%, 14%, 14%) with stratified group splitting so every lesion appears in one split only. Training augmentation consists of random flips, transposition (all eight dihedral orientations) and mild brightness jitter; validation and test images are never augmented.

5.2 Label-budget protocol

To simulate scarce annotation, the labelled training set is subsampled to 10%, 25%, 50%, 100% of the training images, stratified by class (at least two images per class). The validation set used for checkpoint selection is reduced in the same proportion so that no method benefits from a large labelled validation pool. The test set is always complete and is never used for tuning. The number of epochs is scaled for small budgets (up to four times) so that the number of optimiser steps remains comparable.

5.3 Models

- ViT (scratch): patch size 8, embedding width 128, 4 pre-norm transformer blocks, 4 heads, learnable class token and positional embeddings (about 0.8 million parameters), randomly initialised.
- ViT + masked pre-training: the same architecture, first pre-trained on the unlabelled training images by hiding 60% of the patches and reconstructing their normalised pixels (a SimMIM-style objective; Xie et al., 2022; He et al., 2022), then fine-tuned with labels. Labels are never used in pre-training and validation/test images are never seen.
- ViT + masked pre-training + weighted loss: as above, with a class-weighted cross-entropy (inverse class frequency) to counter imbalance.
- CNN baseline: four convolution-batch-norm-ReLU-max-pool blocks with global average pooling (about 0.3 million parameters).

5.4 Training and evaluation

All models are optimised with AdamW (Loshchilov & Hutter, 2019) using a one-cycle learning-rate schedule, weight decay 0.05, batch size 64 and gradient clipping, and are implemented in PyTorch (Paszke et al., 2019). The checkpoint with the best validation macro-F1 is evaluated once on the test set. The primary metric is macro-F1, which weights all classes equally; per-class recall, accuracy and balanced accuracy are reported as supporting measures. Every experiment is repeated over several seeds, which also change the data split, and mean ± standard deviation is reported.

6. Experimental Setup

The exact configuration is inserted automatically after the experiments are run.

7. Results

No results yet. Execute python run_all.py to fill this chapter automatically.

8. Discussion and Limitations

- The study uses small models and low-resolution images to keep experiments reproducible on modest hardware; conclusions may change with larger models, higher resolution or large-scale pre-training.
- Self-supervised pre-training here uses only the unlabelled images of the training split; external unlabelled dermoscopy data could behave differently.
- A single dataset is used; generalisation across devices, hospitals and skin tones is not demonstrated, and skin-tone annotations are not available in HAM10000 (Ansari et al., 2024).
- A small number of seeds limits statistical power; differences within one standard deviation should not be over-interpreted.
- The prototype is not a medical device. Any clinical use would need external validation, regulatory approval and human oversight.

9. Conclusion

Conclusions are generated from the measured results after the experiments have been run.

10. Reproducibility

The complete code, configuration and instructions are in the project folder. python run_all.py --dataset synthetic runs the pipeline end-to-end without downloads; python run_all.py --dataset ham10000 --data-root path/to/HAM10000 reproduces the real-data study. All figures, tables, the report, the PDF and the slides are produced from results/results.json, so documents always match the measured numbers.

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